

A PROJECT TITLE
ON
HIGH-DIMENSIONAL NETWORK LOG VISUALIZATION USING PCA and t-SNE

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: **HIGH-DIMENSIONAL NETWORK LOG VISUALIZATION USING PCA and t-SNE**

1. Abstract:

The rapid growth of digital communication and cloud computing has led to an enormous increase in the volume and complexity of network log data, making it difficult to analyze using conventional visualization techniques. This project, "**High-Dimensional Network Log Visualization Using PCA and t-SNE,**" presents an efficient approach for analyzing high-dimensional network log data through dimensionality reduction and visualization. The proposed system employs **Principal Component Analysis (PCA)** to reduce the number of features while preserving the maximum variance in the dataset, followed by **t-Distributed Stochastic Neighbor Embedding (t-SNE)** to capture complex non-linear relationships and generate meaningful two-dimensional visualizations. These techniques enable the identification of hidden patterns, clusters, and anomalies within network traffic, assisting network administrators and cybersecurity professionals in detecting suspicious activities, performance issues, and potential security threats. By transforming complex multidimensional data into an intuitive visual format, the system improves data interpretation, supports exploratory analysis, and enhances decision-making in network monitoring and security management. The proposed approach not only simplifies the analysis of large-scale network logs but also retains essential information required for accurate pattern recognition and anomaly detection. Overall, this project demonstrates that integrating PCA and t-SNE provides an effective and scalable solution for visualizing high-dimensional network log data, enabling faster insights, improved network analysis, and more reliable identification of abnormal behavior in modern network environments.

2. INTRODUCTION:

The rapid growth of computer networks has resulted in the generation of massive amounts of network log data containing multiple features such as IP addresses, protocols, timestamps, packet sizes, and traffic information. Analyzing this high-dimensional data using traditional visualization techniques is difficult because the large number of features makes it challenging to identify meaningful patterns and anomalies. According to the Machine Learning course, dimensionality reduction techniques such as **Principal Component Analysis (PCA)** and **t-Distributed Stochastic Neighbor Embedding (t-SNE)** are widely used to simplify high-dimensional datasets while preserving important information for visualization.

In this project, PCA is used to reduce the dimensionality of network log data by retaining the maximum variance present in the dataset. The reduced data is then visualized using t-SNE, which preserves the relationships between similar data points and represents them in a two-dimensional space. The resulting visualizations help identify clusters, hidden patterns, and abnormal network behavior, making network monitoring and cybersecurity analysis more efficient. Thus, the proposed system provides an effective solution for understanding complex network log data through dimensionality reduction and visualization.

3. SOFTWARE REQUIREMENTS:

Operating System

- Windows 10/11 or Ubuntu Linux

Programming Language

- Python 3.10 or above

Development Environment

- Jupyter Notebook / Visual Studio Code

Python Libraries

- NumPy
- Pandas
- Scikit-learn
- Matplotlib
- Seaborn

Machine Learning Techniques

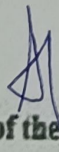
- Principal Component Analysis (PCA)
- t-Distributed Stochastic Neighbor Embedding (t-SNE)

Dataset

- Network Log Dataset (CSV format)

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB
- Storage: 2 GB free space

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Signature of the Guide

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Course Instructor